

ADITYA PATIL

B.Tech Information Technology | NMIMS Shirpur | Expected Graduation: 2027

Project Summary for FOX Solutions Pvt. Ltd. — Special Engineer Trainee (Software Development)

CivicConnect — Civic Issue Reporting & Resolution Platform

Project overview: CivicConnect is a full-stack platform designed to help citizens report civic issues and enable administrators and field workers to track and manage issue resolution. The project focuses on multi-role workflows, backend API development, data handling, and real-time status updates.

Technology Stack

Backend	Databases	Integrations
Node.js, Express.js, REST APIs	MongoDB, MySQL	Socket.IO, Gemini API, Google Maps

Key Features & Contributions

- **REST API development:** Built backend services to support issue reporting, status tracking, and workflows for citizens, administrators, and field workers.
- **Data management:** Integrated MongoDB and MySQL to support application data and structured querying.
- **Authentication and authorization:** Implemented role-based access control (RBAC) to separate permissions across the three user roles.
- **Real-time updates:** Used Socket.IO to support live status tracking and communication within the application.
- **Issue analysis:** Implemented geospatial duplicate detection using a 100-metre radius and Gemini API-based complaint categorization.
- **Validation and debugging:** Applied server-side validation and worked across API, database, and application workflows to support reliable operation.

Engineering Skills Demonstrated

Backend application development • API design and integration • Database connectivity and querying • Authentication and access control • Real-time application workflows • Debugging and validation • Working with third-party services

Relevance to FOX Solutions

This project demonstrates practical experience in designing and developing application components, writing and debugging code, integrating databases and APIs, and implementing role-based business workflows. These skills align with the Software Development trainee role described in the FOX Solutions campus drive. I am interested in applying these foundations to production software and continuing to learn within an engineering team.

Candidate: Aditya Patil | Program: B.Tech Information Technology, NMIMS Shirpur | Graduation: 2027